FIT Lecture # 5

Engr. Naveera Sami

- All computer systems consist of two major components, namely, hardware and software. The hardware refers to the physical equipments that are necessary for performing various operations, such as reading and processing data, storing results and providing output to the users in a desired form. The software refers to a set of computer programs that are required to enable the hardware to work and perform these operations effectively.
- A computer program is basically a set of logical instructions, written in a computer programming language that tells the computer how to accomplish a task. The software is therefore an essential interface between the hardware and the user

TYPES OF COMPUTER SOFTWARE

As stated earlier, a computer software performs two distinctive tasks. The first task is to control and coordinate the hardware components and manage their performances and the second one is to enable the users to accomplish their required tasks. The software that is used to achieve the fi rst task is known as the system software and the software that is used to achieve the second task is known as the application software. While the system software is essential for a computer to work, the application software is the additional software required for the user to perform a specifi c job. The system software not only controls the hardware functions but also enables the hardware to interact with the application software as well as the users

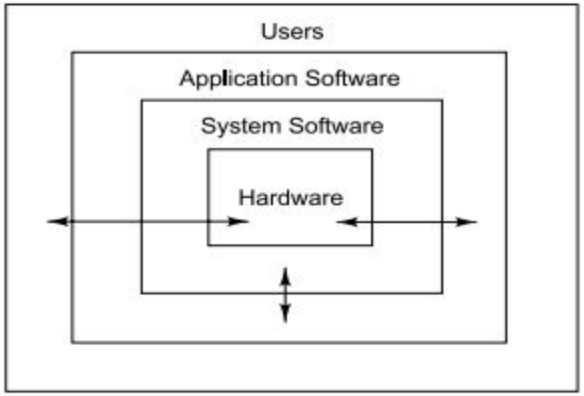

Fig. 10.2 Layers of software and their interactions

1. System Software
System software consists of many different programs that manage and support different tasks. Depending upon the task performed, the system software can be classified into two major groups:

System management programs used for managing both the hardware and software systems.
System development programs used for developing and executing application software.

2. Application Software
Application software includes a variety of programs that are designed to meet the information processing needs of end users. They can be broadly classified into two groups:
Standard application programs that are designed for performing common application jobs.
Unique application programs that are developed by the users themselves to support their specific needs.

Computer Software

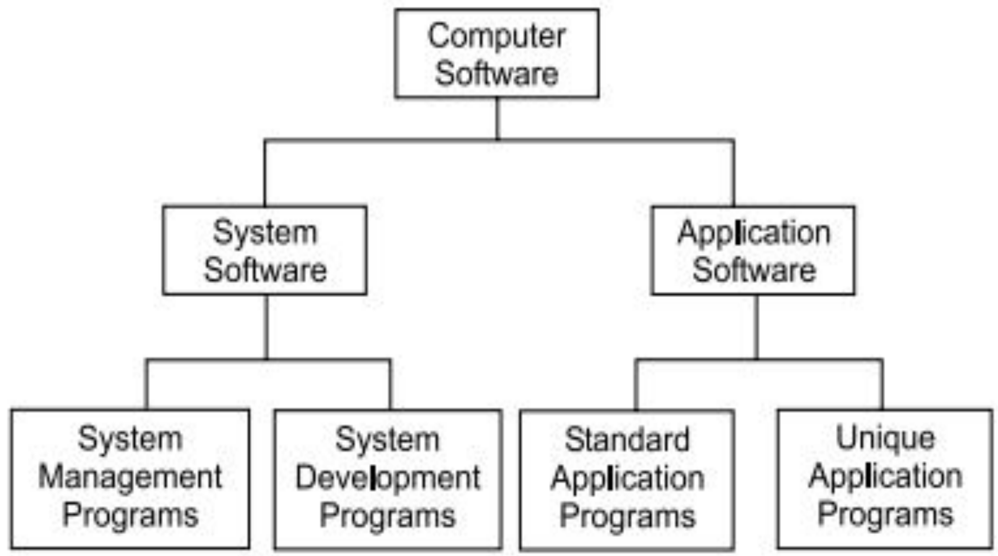

Fig. 10.3 Major categories of computer software

SYSTEM MANAGEMENT PROGRAMS:

System management programs are those programs that are meant for operating the hardware system and managing their resources effectively. They also enable the users to perform certain utility functions, such as creating backup files,recovering damaged fi les and merging files.They minimise the human intervention during processing and aid in maximising the productivity of a computer system. System management programs include:

Operating system

Utility programs

Device drivers

These programs work in close interaction with each other

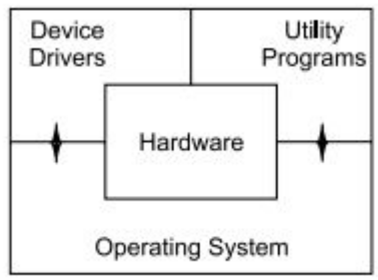

Fig. 10.4 System Management Programs

Operating System

Operating System (OS) is the principal component of system software and is responsible for overall management of computer resources. It also provides an interface between the computer and the user and helps in implementing the application programs.

Major functions of an operating system are:

1. Scheduling and execution of all processes.
2. Allocation and management of main memory and other storage areas to the programs.
3. Coordination and assignment of different hardware devices to the programs.
4. Creation, storage and manipulation of files required by the various processes.

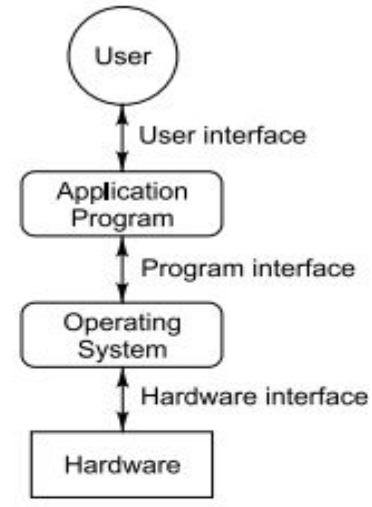

Fig. 10.5 Hardware-OS-User interface

Operating System:

5. Determining and maintaining the order of execution of programs.
6. Interpretation of commands and instructions.
7. Coordination and assignment of other development and utility programs.
8. Providing a friendly interface between the computer and the user.
9. Ensuring security of access to computer resources.

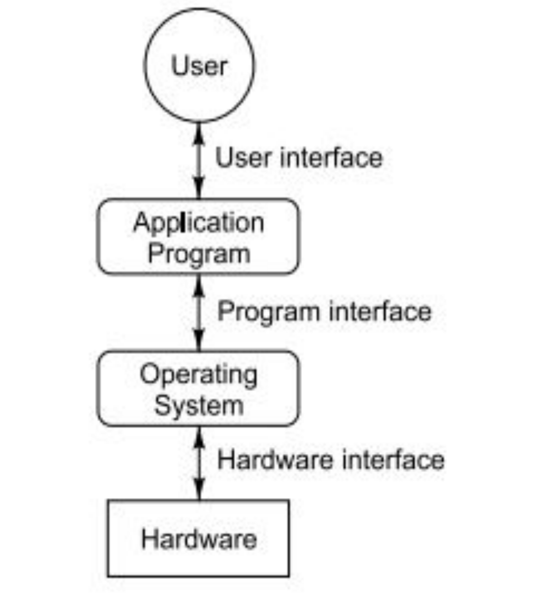

Fig. 10.5 Hardware-OS-User interface

Utility Programs:

Utility programs refer to small programs, which provide additional capabilities to the computer system in addition to the ones provided by the operating system. They enable an operating system to perform some additional tasks, such as searching and printing the fi les and scanning the viruses, etc. A utility program is not an essential part of an operating system, because it does not help the operating system in the execution of a command or a program. A utility program only provides the additional features to the computer system. In other words, an operating system can execute most of the programs without having the utility programs. Utility programs are added to an operating system to perform many different tasks, that include:

Utility Programs:

- Search and replace. It enables the operating system to search a fi le on the basis of the specified search criteria.
- Print. It enables an operating system to initiate the print operation of the printer connected with the computer system.
- Disk defragmenter. It helps in defragmenting the memory space. Defragmentation is the process of storing the data at a single place in the memory instead of disjointed memory locations.
- System profiler. It provides the information related to the various hardware and software components installed in the computer system. In other words, it provides a list of the hardware components, which are presently connected with the computer system, and the software components, which are currently installed in the computer system.
- Encryption. It enables the operating system to generate the encrypted format of the messages or the files, which have to be transmitted from one system to another system over the network or the Internet.

Utility Programs:

- Virus scanner. It enables the operating system to detect viruses and bugs, which may affect the

correct functioning of the computer system.

- Backup. It enables the creation of a copy of various fi les and folders on a secondary medium such as magnetic disk and magnetic tape, in order to keep the original data safe.
- Data recovery. It enables the retrieval of lost data from a corrupted or damaged primary storage medium.

Like operating systems, utility programs are prewritten by manufacturers and supplied with the hardware. They may also be obtained from standard software vendors. A good range of utility programs can make the life much easier for the user.

Device Drivers:

A computer system is connected with multiple input and output (I/O) devices, so that it can communicate with the end user. In order to interact with the I/O devices, the computer system requires special software called device driver. The device driver acts as a translator between the I/O devices and the computer. A device driver of the input device interprets the input provided by the user into the computer
understandable form and directs it to the operating system. Similarly, the device drivers of the output devices translate the output generated by the computer into the user understandable format and display it on the screen. In other words, a device driver is special software that enables a hardware device, such as keyboard, monitor and printer to perform an operation according to the command given by the end user.

Device Drivers:

Whenever a program needs to use a connected hardware device, it issues a command to the operating system. The operating system calls the respective routine or process of the device driver, which instructs the device to perform the task. For providing instruction to the devices, the device driver sends various control and data signals through buses and cables to the device. In order to acknowledge the device driver's signal, the device sends the acknowledgment signals to the device driver. After performing the task, the device sends a message to the device driver, which returns the value to the calling program of the operating system. In other words, a device driver instructs a hardware device the way it should accept the input from the operating system and the way in which it should communicate with the other units of the computer system.

Figure illustrates the working of the device driver.

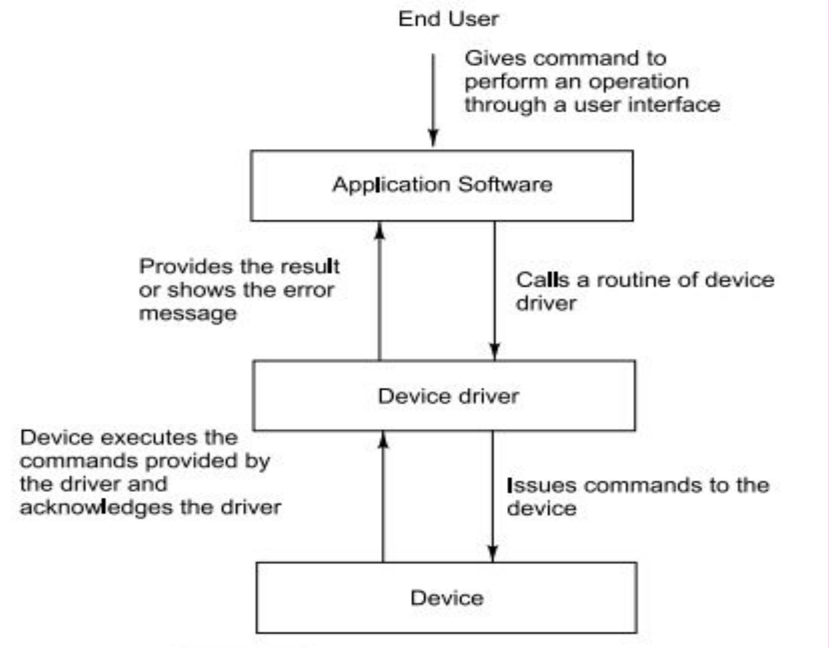

Fig. 10.6 Working of a device driver

SYSTEM DEVELOPMENT PROGRAMS

System development programs known as programming software allow the users to develop programs in different programming languages. The process of developing and executing a program involves the following tasks:

1. Debugging the program
2. Linking the various variables and objects with the libraries files
3. Translating the code from one language to another
4. Running the machine code to perform the desired task

In order to carry out these tasks, we need the following system development tools:

Language translators, Linkers, Debuggers, Editors

Language Translators

Language translator is used to convert the program code written in one language to another language. The program code provided as an input to the language translator is known as source code. The source code is a high level or an assembly language program. The language translator converts the high-level language program into the low-level language program called object code. Compiler, interpreter and assembler are the most common examples of language translator. A compiler translates a high-level program into a low-level program and an assembler translates an assembly language program into a low-level program. An interpreter also produces a low-level program from a high-level program, but the working of the interpreter is not similar to that of the compiler.
An interpreter processes the high-level program line-by-line and simultaneously, produces the low-level program.

On the other hand,
a compiler compiles the high-level program in one go.
Figure shows
the input and the output
of the various language translators.

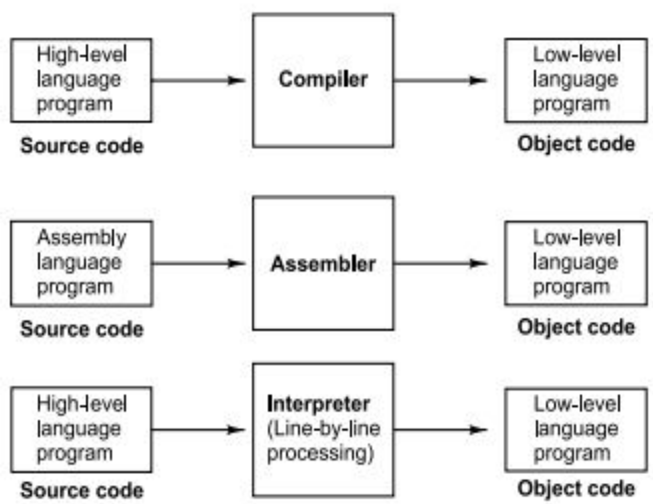

Fig. 10.7 Language translators

Linkers

Most of the high-level languages allow the developer to develop a large program containing multiple modules. Linker arranges the object code of all the modules that have been generated by the language translator into a single program. The execution unit of the computer system is incapable of linking all the modules at the execution time and therefore, linker is regarded as one of the important software because of its ability to combine all the modules into a single program. Linker assembles the various objects generated by the compiler in such a manner that all the objects are accepted as a single program during execution. Linker also includes the links of various objects, which are defi ned in the runtime libraries. In many cases, linker inserts the symbolic address of the objects in place of their real address. Figure 10.8 illustrates the working of a linker.

Debuggers

Debugger is the software that is used to detect the errors and bugs present in the programs. The debugger locates the position of the errors in the program code with the help of what is known as the Instruction Set Simulator (ISS) technique. ISS is capable of stopping the execution of a program at the point where an erroneous statement is encountered. Debugger is divided into two types, namely, machine-level debugger and symbolic debugger. The machine-level debugger debugs the object code of the program and shows all the lines where bugs are detected. On the other hand, the symbolic debugger debugs the original code, i.e., the high-level language code of the program. It shows the position of the bug in the original code of the program developed by the programm

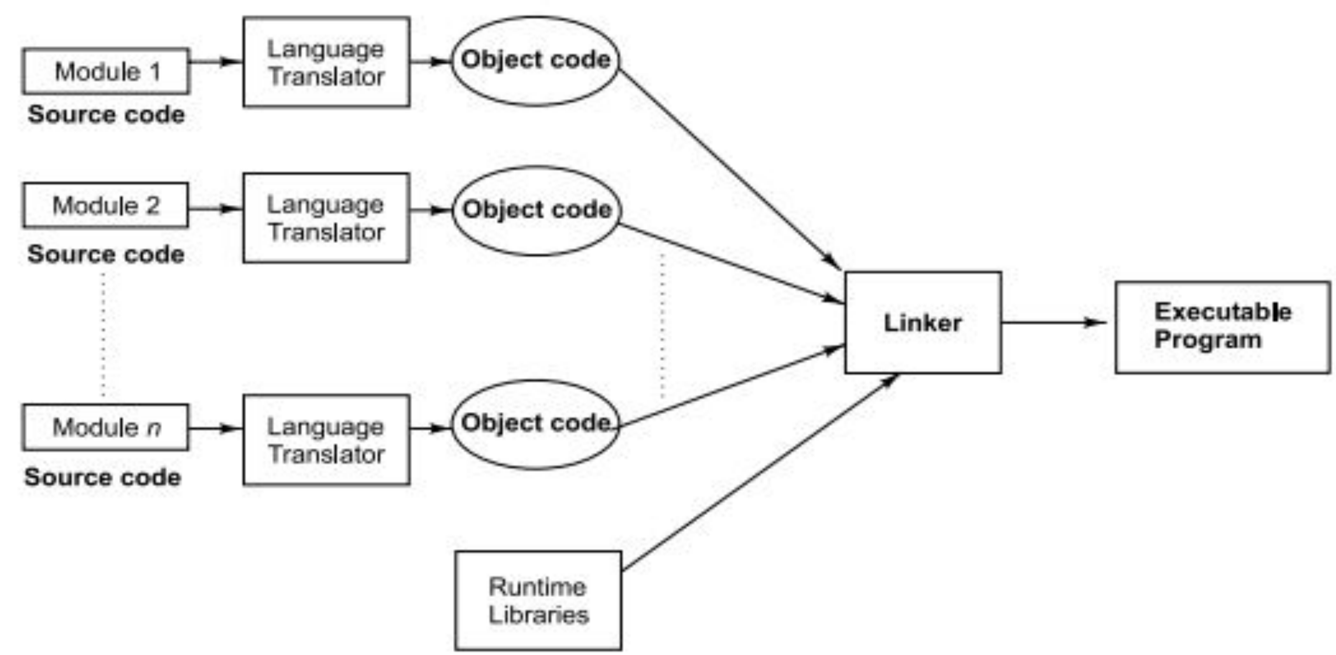

Fig. 10.8 Working of linker

Debuggers

While debugging a program, the debugger performs a number of functions other than debugging, such as inserting breakpoints in the original code, tracking the value of specific variables, etc. In order to debug the program, a debugger helps us to perform the following tasks:

1. Step-by-step execution of a program
2. Back tracking for checking the previous steps
3. Stopping the execution of the program until the errors are corrected

Editors

Editor is a special program that allows the user to work with text in the computer system. It is used for the documentation purposes and enables us to edit the information present in an existing document or a file. The editor enables us to perform the various editing operations such as copy, cut and paste while editing the text. On the basis of the content edited by the editors, they are divided into the following categories:

1. Text editor It is used to edit plain text. An operating system always includes a text editor for updating the configuration files.
2. Digital audio editor It is used to edit the information related to the audio components of a multimedia application. These editors are used in audio applications where editing the music and the sound signals is necessary.
3. Graphics editor It is used to edit the information related to the graphical object. These editors are generally used in the multimedia applications where the user is working with multiple animation objects.
4. Binary file editor It is used to edit the digital data or the binary data, i.e., data having strings of 0s and 1s.
5. HTML editor It is used to edit the information included in the web pages.
6. Source code editor It is used to edit the source code of a program written in a programming
language such as C, C++ and Java.

STANDARD APPLICATION PROGRAMS

Standard application programs, also known as general-purpose application programs, are programs that perform certain common information processing tasks for the users. For example, preparing documents such as letters and notes are common among almost all organisations as well as individuals. Similarly, creating and managing databases of products and employees are very common activities in large organisations. Ready-to-use programs for such applications are available from hardware or software vendors. These programs would not only increase significantly the productivity of users but also decrease the investment in terms of time and cost. Examples of standard application programs include:

Word processor

Spreadsheet

Database Manager

Desktop Publisher

Web Browser

We shall briefly discuss some of these programs in this section.

Word Processor

Word processor is an application software that is generally used to create documents such as letters, reports, etc. It is one of the most commonly used computer applications. Word processor enables us to enter text in a document in a particular format. It also allows us to modify the existing text and apply the formatting features, such as boldface and italic, change the colour of the text, change the writing style of the text, etc. The word processor application software also allows us to perform different operations, such as cut-paste for moving the text within the document, copy-paste for copying the text within a document or from one document to another document, etc. MS Word is the most common example of a word processor. Figure shows the default window of MS Word.

Fig. 10.9 The default window of MS Word

10.5.2 Spreadsheet

Spreadsheet is an application software that enables the user to maintain a worksheet in which information is stored in different cells. Each cell in the spreadsheet can contain numeric and alphanumeric data. Spreadsheet is also used as a computational tool by applying formulas on the values of cells. If we change the value of a cell in a spreadsheet, then all the values, which are dependent on that value, get automatically updated according to the new value. As a result, the users, who need to store and maintain financial information, always prefer a spreadsheet for storing the information.

MS Excel is the most common example of the spreadsheet software. Figure 10.10 shows the default window of MS Excel.

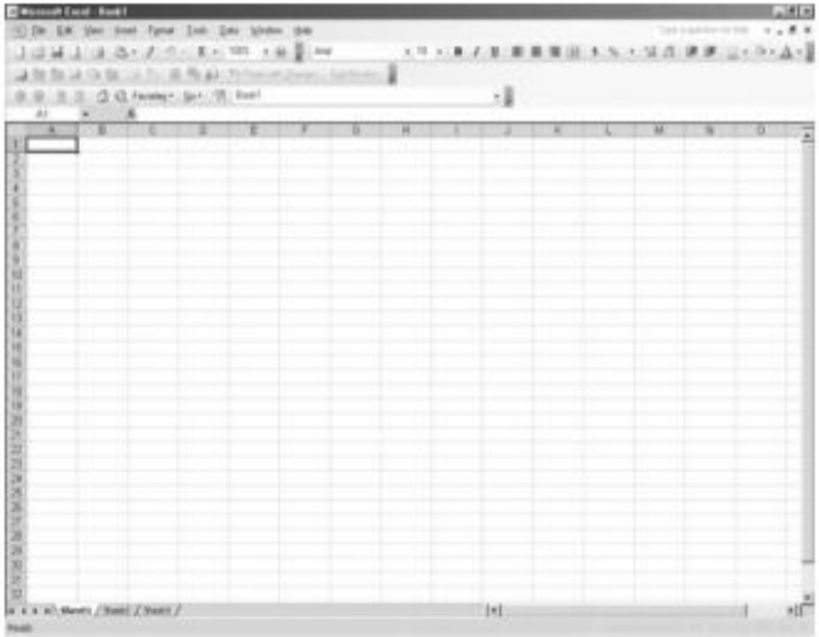

Fig. 10.10 The default window of MS Excel

10.5.3 Database Management System

Database Management System (DBMS) is a computer software that helps us to store and maintain records in a database. Database refers to a set of records, which are stored in a structured manner in the computer system. Record refers to a set of similar or dissimilar values related to an entity such as student and employee. DBMS allows a user to perform various operations such as search, update and delete on the data stored in the database. With the help of DBMS an end user can store, retrieve, and modify the data in an easy manner. The advantages of DBMS are as follows:

- It enables the user to organise all the information in a strategic manner.
- It helps the users in maintaining consistency in the information stored in the database.
- It helps the user in maintaining data integrity while performing the various operations on the database values.
- It allows the user to maintain the security of the confidential information by password-protecting the database.
- It helps in accessing information from any system connected in a networked system environment.

The DBMS used for maintaining a database always uses specific data model such as relational, hierarchical and network that describes the physical and the logical arrangement of data in the database. In the relational data model, different records are arranged in the form of tables containing rows and columns. In the hierarchical model, records are arranged in the form of a tree where each record is attached with one parent node and one or more child nodes. In the network model, data is arranged in the form of a network where each node is connected with one or more parent nodes as well as child nodes. Figure 10.11 shows the arrangement of records in different data models of the DBMS.

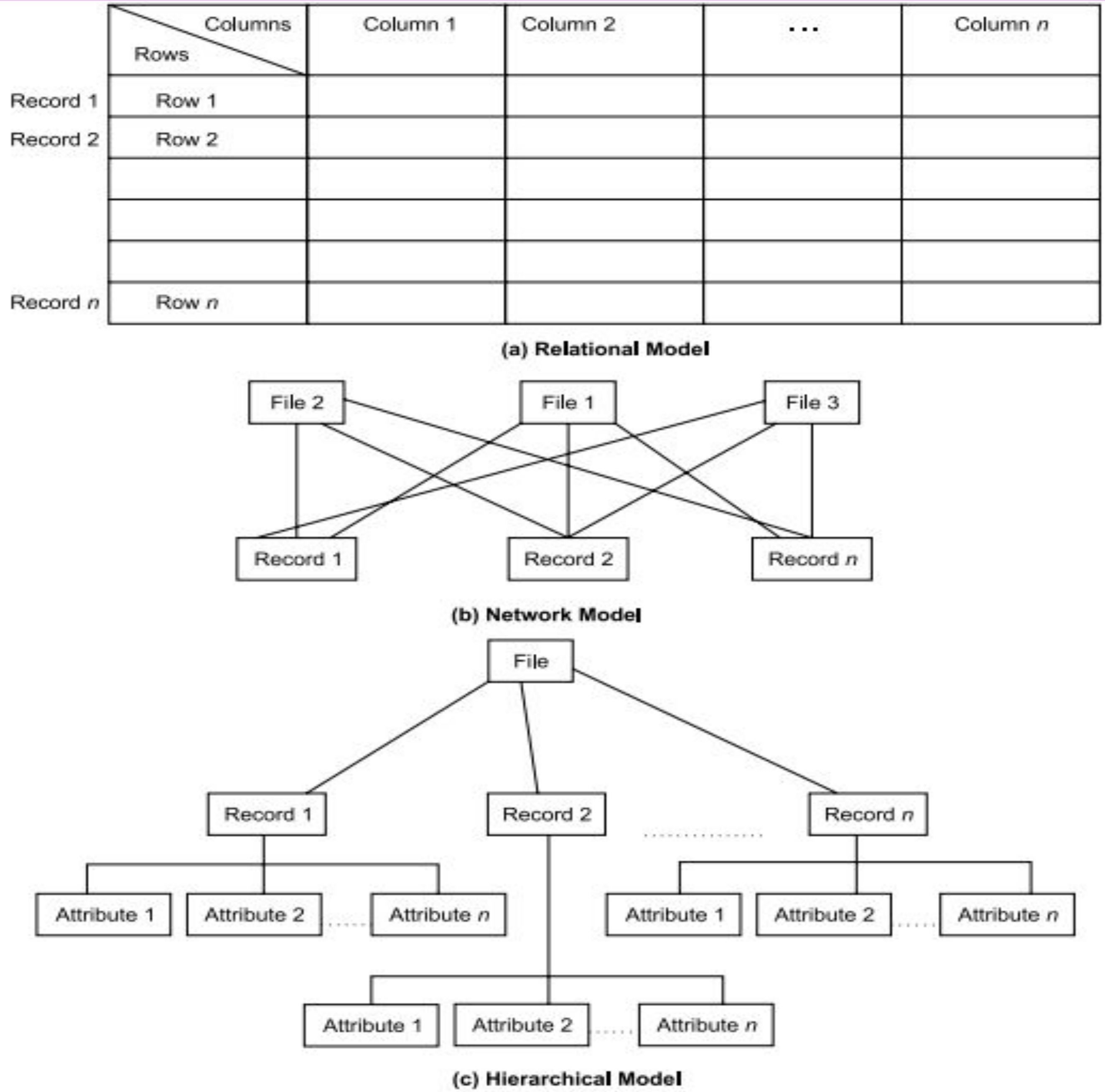

Fig. 10.11 Different data models used to design DBMS

Desktop Publishing System

Desktop publishing system enables the user to perform various activities required for publishing of a page or set of pages. These activities include organising the content layout by setting margins and justification properties of the page.

In the desktop publishing system, a page layout software is installed on a computer system that enables the end users to view the changes done at the time of designing the document. The page layout software uses the ‘ What You See Is What You Get (WYSIWYG)’ technique for displaying the preview of the page being designed. Desktop publishing system is also referred as DTP.
Desktop publishing system is generally used in the publishing industry, where there is a need to publish information either through print media or display media. It helps the users to design the desired page layout containing different components such as text, graphics and images. Some of the commonly used DTP application software include: Quark XPress, InDesign, Microsoft Publisher, Apple Pages.

UNIQUE APPLICATION PROGRAMS

There are situations where organisations need to develop their own programs to accomplish certain tasks that are unique to their areas of operations. Similarly, individuals like scientists, engineers, accountants, teachers and other professionals write their own programs to solve their problems. Such programs are known as unique application programs or application-specific programs. These programs are also referred to as end-user application programs. Examples of unique applications include:

Managing the inventory of a store

Preparing pay-bills of employees in an organisation

Processing examination results of students

Reserving seats in trains or airlines

Analysing mathematical models

Computing personal income tax

And many more

These programs are usually developed in one of the high-level languages, such as C, C++ or Java and therefore, developing such programs in-house would require skilled programmers with deep knowledge not only in programming languages but also in programming environment.

We will discuss below in this section two applications which are usually developed in-house to suit the user's unique requirements.

Inventory Management System

Inventory management system helps in keeping a record of the huge amount of items stored in warehouse and storerooms. In order to keep a record of inventory or items, application software called inventory management system, which makes use of DBMS, is used. DBMS is responsible for storing the information related to the items in the form of various fi les and tables. It also helps in storing the information related

to the location, where a particular item is placed. Inventory management system enables a user to get a report on the items stored in the warehouse or storeroom on the basis of the different conditions, such as availability of items, shipping date of item, etc. The database of the inventory management system identifies each item by a unique identification number. The identification number is used as the primary key of an item in the database.

Inventory Management System

Primary key refers to an attribute, which must contain a unique value for each item of the database. The identification number can be dependent on the various properties, such as type, model and manufacturing date of the items, etc. Nowadays, the identification number of an item is specified on the barcodes printed on each item. Barcode is a group of small and black vertical lines of varying widths. The barcodes are not understandable by a human being and is always read by an input device called barcode reader. A barcode reader is similar to the handheld scanner that scans the barcode of the item in order to enter the barcode in the database. Figure shows the sample barcode of an item. The use of barcode technology with inventory management system saves a lot of time as the user is not required to manually enter the item details. These details are automatically read by the barcode scanner and stored in the database of the inventory management system. An inventory management system finds its application in those organisations where a constant maintenance and updation of inventory information is required.

ISBN 000-867-5309

9 780008 675301

Fig. 10.12 Barcode of an item

Pay-roll System

Pay-roll system is the application software that enables us to perform the various activities related to payroll accounting and pay-roll administration process. It is used by the HR department of an organisation for the purpose of computation of salary of employees. A typical pay-roll processing system provides an interface to the user through which the user can enter salary related details, such as attendance, tax deduction, etc.

A pay-roll system is an effective and efficient way of computing the salary of employees as it eliminates the manual task of performing computations and automates most of the tasks related to pay-roll processing.

A pay-roll system can be custom built or it can be an off-the-shelf application software. Several ERP products include a dedicated module on pay-roll processing which is particularly used in enterprise-wise implementation of pay-roll processing.

A pay-roll system can also be used to print or e-mail the salary slips of employees.


THANK
YOU